

Adding Rational Numbers

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A **rational number** is any number that can be written in the form

$$\frac{a}{b},$$

where a and b are integers and $b \neq 0$.

Adding Fractions

To add **fractions** with the same denominator, add the numerators and keep the denominator.

$$\frac{3}{7} + \frac{2}{7} = \frac{3 + 2}{7} = \frac{5}{7}$$

Different Denominators

To add **fractions** with different denominators, first find the **least common denominator (LCD)**, then add.

$$\frac{2}{3} + \frac{1}{6} = \frac{4}{6} + \frac{1}{6} = \frac{5}{6}$$

Forms of Rational Numbers

Rational numbers may be written as:

- ▶ **fractions**

- ▶ **integers**

- ▶ **decimals**

Examples

Example 1

Find $8 + \frac{5}{3}$.

$$8 = \frac{24}{3}$$

$$\frac{24}{3} + \frac{5}{3} = \frac{29}{3} = 9\frac{2}{3}$$

Example 2

Find $1.2 + \frac{4}{5}$.

$$\frac{4}{5} = 0.8$$

$$1.2 + 0.8 = 2$$

Example 3

Find $\frac{5}{4} + \frac{7}{8}$.

$$\frac{5}{4} = \frac{10}{8}$$

$$\frac{10}{8} + \frac{7}{8} = \frac{17}{8} = 2\frac{1}{8}$$

Practice Exercises

Practice 1

Solve completely.

$$8 + \frac{5}{3}$$

Practice 2

Solve completely.

$$\frac{4}{3} + 2$$

Practice 3

Solve completely.

$$1.2 + \frac{4}{5}$$

Practice 4

Solve completely.

$$\frac{3}{7} + \frac{2}{7}$$

Practice 5

Solve completely.

$$\frac{5}{8} + \frac{3}{8}$$

Practice 6

Solve completely.

$$\frac{2}{3} + \frac{1}{6}$$

Practice 7

Solve completely.

$$\frac{5}{4} + \frac{7}{8}$$

Practice 8

Solve completely.

$$2.5 + 1.8$$

Practice 9

Solve completely.

$$3.75 + 2.4$$

Practice 10

Solve completely.

$$\frac{2}{5} + 1\frac{1}{10}$$

Activities

Activity 1

Solve the problem. Show your solution.

$$6 + \frac{4}{3}$$

Activity 2

Solve the problem. Show your solution.

$$\frac{5}{3} + 3$$

Activity 3

Solve the problem. Show your solution.

$$1.4 + \frac{3}{5}$$

Activity 4

Solve the problem. Show your solution.

$$\frac{4}{9} + \frac{3}{9}$$

Activity 5

Solve the problem. Show your solution.

$$\frac{7}{10} + \frac{2}{10}$$

Activity 6

Solve the problem. Show your solution.

$$\frac{5}{8} + \frac{1}{4}$$

Activity 7

Solve the problem. Show your solution.

$$\frac{3}{2} + \frac{5}{6}$$

Activity 8

Solve the problem. Show your solution.

$$4.2 + 2.7$$

Activity 9

Solve the problem. Show your solution.

$$6.35 + 1.8$$

Activity 10

Solve the problem. Show your solution.

$$\frac{3}{4} + 1\frac{1}{2}$$

Thank you!